POWER PLANT ENGINEERING

Subject Code: 23A03702e

UNIT I: Sources of Energy, Power Plant Economics and Environment

I. Introduction to Sources of Energy

- ***Energy*** = capacity to do work. Power = rate of doing work (kW, MW). Energy = power x time (kWh).
- ***Power plant*** = an assembly of systems / equipment that converts a primary (natural) energy source into ***electrical energy*** on a large scale.
- Primary sources: fuels (coal, oil, gas), water, wind, sun, nuclear fuel, biomass, tides, geothermal.
- Electrical energy is a ***secondary energy*** -- easy to transmit, clean at the point of use, easy to convert to light / heat / motion.

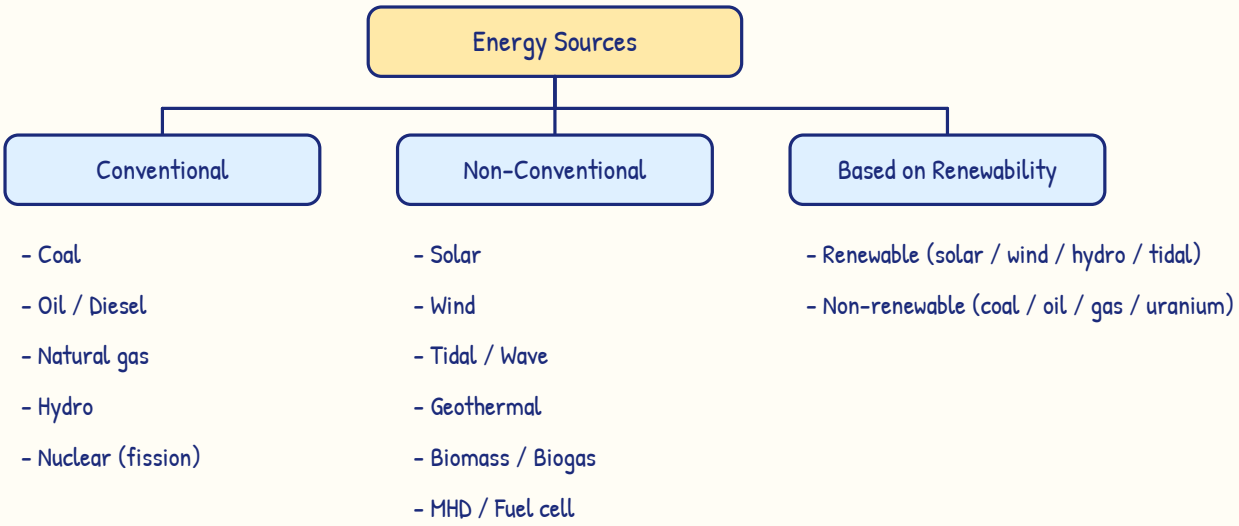


Fig: Classification of energy sources

Conventional vs Non-conventional sources

Point	Conventional	Non-conventional
Nature	Mostly exhaustible (fossil, nuclear)	Mostly renewable, inexhaustible
Examples	Coal, oil, gas, hydro, nuclear	Solar, wind, tidal, biomass, geothermal
Capacity	Large-scale, base-load plants	Small / medium, scattered, intermittent
Pollution	High (CO_2, SO_x, NO_x, ash)	Little or none

Point	Conventional	Non-conventional
Capital cost	Medium to high	Generally high per kW (low load factor)
Running cost	Fuel cost is large	Fuel is free; maintenance only

Why non-conventional sources?

- Fossil fuel reserves are **limited** and costly; import of crude oil burdens the economy.
- Environmental problems: global warming, acid rain, smog.
- Suitable for remote villages where grid extension is uneconomical.

2. Resources and Development of Power in India

- India's installed capacity is about **4.5 lakh MW** (approx.), with **coal / lignite ~ 50%** of it; renewables have the fastest growth.
- **Coal**: reserves mainly in Jharkhand, Odisha, Chhattisgarh, West Bengal, MP -- mostly **non-coking, high-ash (30-45%), low sulphur**, GCV about 3500-5000 kcal/kg.
- **Oil and gas**: Mumbai High, Assam, Krishna-Godavari basin, Gujarat; part imported as LNG.
- **Hydro potential**: about 1,45,000 MW (at 60% load factor); Himalayan rivers, Western Ghats; developed share is only about one-third.
- **Nuclear**: uranium (Jharkhand, Andhra), **thorium** (Kerala monazite sands -- among the largest in the world). Plants: Tarapur, Kalpakkam, Kaiga, Kudankulam, Rawatbhata.
- **Solar**: 300 sunny days a year, 4-7 kWh/m^2/day (Rajasthan, Gujarat). **Wind**: Tamil Nadu, Gujarat, Karnataka, Maharashtra.

Development of power (milestones)

- 1897 Darjeeling hydro station; 1902 Sivasamudram (Karnataka) -- first major hydro plant in India.
- 1948 Electricity (Supply) Act; 1950s: Five Year Plans -- State Electricity Boards (SEBs) formed.
- 1975 NTPC and NHPC; 1969 **Tarapur** nuclear unit; 1989 Power Grid Corporation (PGCIL) -- **National Grid**.

- 2003 Electricity Act (competition, open access); today a ***One Nation, One Grid*** system with 5 regional grids synchronised.
* Power sector is run by Centre (NTPC, NHPC, NPCIL, PGCIL), States (GENCOs, DISCOMs) and the private sector.

3. Power Plant Economics

- A good power plant must supply ***reliable, cheap and sufficient*** power. Economics decides the type of plant, size, and location.

3.1 Capital cost

- Initial investment for land, buildings, boiler / turbine / generator, civil works, transmission, installation and ***interest during construction***.
- Typical cost per kW: Hydro (highest, dams) > Nuclear > Steam > Gas turbine > Diesel (lowest).

3.2 Investment (fixed) charges

- Charges that are ***independent of units generated*** (paid even if the plant is idle):
  - ***Interest*** on capital (the biggest part, say 8–12%)
  - ***Depreciation*** (loss of value with time)
  - Taxes and insurance
  - Salaries of permanent staff, rent

Depreciation methods

- ***Straight line:*** annual depreciation = (P − S) / n, with P = first cost, S = salvage value, n = life in years.
- ***Diminishing balance:*** fixed % of the book value each year.
- ***Sinking fund:*** equal annual deposit q so that it accumulates to (P − S) at interest i: $q = (P - S) \times i / ((1 + i)^n - 1)$

3.3 Operating costs

- ***Fuel cost*** (largest for thermal plants), operating labour, lubricants and water, maintenance and repairs, supplies. They vary with the ***units generated***.
* Total cost per year = Fixed cost (Rs/kW year) + Running cost (Rs/kWh) x Energy generated

4. General Arrangement of Power Distribution

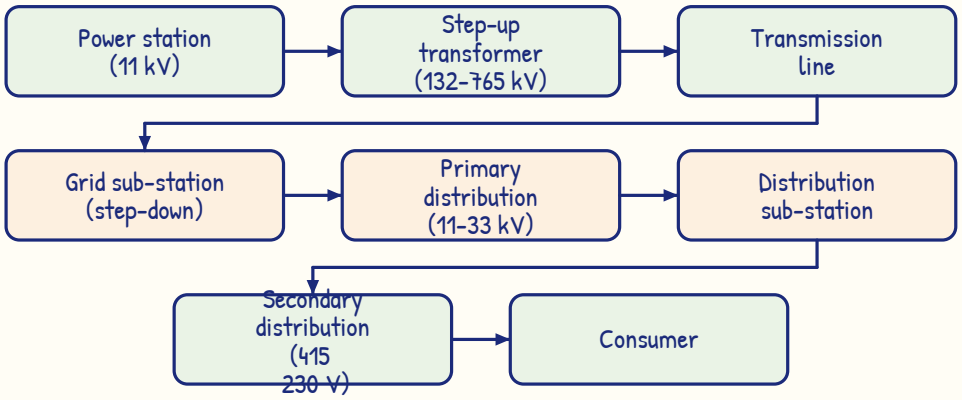


Fig: Path of electric power from station to consumer

- **Generation** at 11 / 13.8 / 21 kV -> stepped up to reduce current and I^2R **loss** during long transmission.
- ***Primary transmission*** 220 / 400 / 765 kV; ***secondary transmission*** 66 / 132 kV; ***primary distribution*** 11 / 33 kV; ***secondary distribution*** 3-phase 415 V, single-phase 230 V.
- Distribution systems: ***radial*** (simple, cheap, poor reliability), ***ring main*** (feeds from two sides, more reliable) and ***interconnected / grid*** (best reliability, common in cities).

5. Load Curves

- ***Load curve*** = graph of load (kW / MW) against time for a given period (day, month or year).
- Shows: peak load, base load, variation, and (from the area) energy generated.

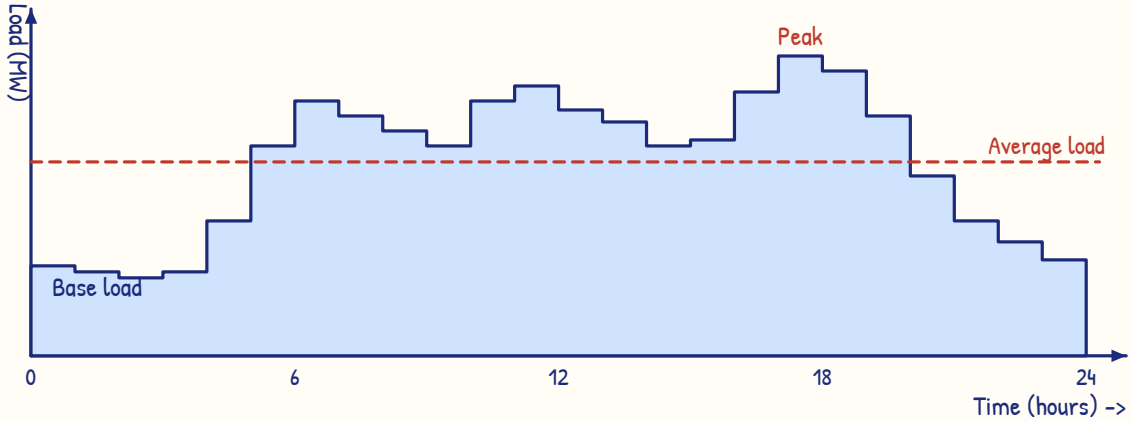


Fig: Daily load curve (chronological)

- Area under daily load curve = ***units generated per day*** (kWh).

- Average load = area / 24 hours. Peak load is usually at evening (lighting + domestic) in India; also a morning peak.

Uses of load curve

- Select the ***number and size of generating units*** and decide on their operating schedule.
- Estimate cost of generation; plan ***maintenance*** during off-peak hours.
- Decide ***base-load and peak-load plants*** (e.g. nuclear / coal = base; hydro / gas turbine / diesel = peak).

Load Duration Curve (LDC)

- Loads are ***rearranged in descending order*** of magnitude, against the number of hours (that load persists). It is always a falling curve.

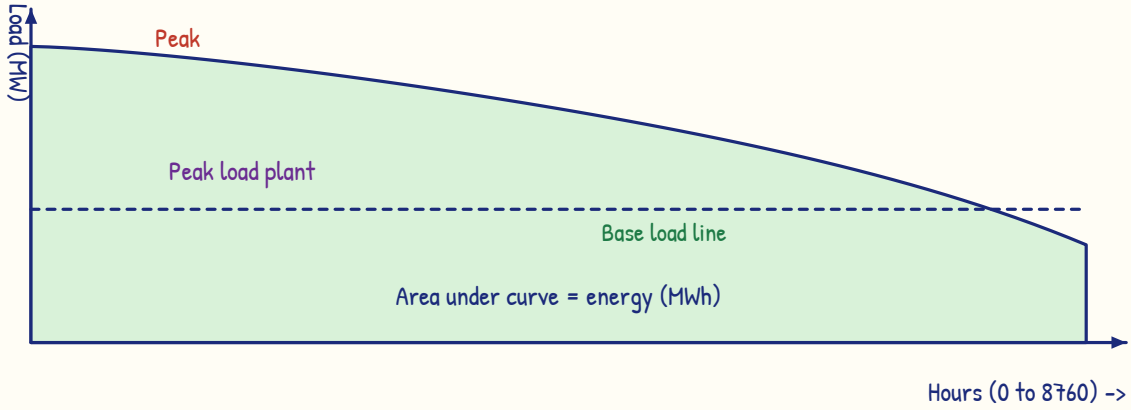


Fig: Load duration curve

- Area under the LDC = ***total energy generated*** (same as the area under the chronological load curve).
- Helps in deciding the ***sharing of load*** between base and peak plants, and in finding capacity factor and load factor easily.

* Chronological load curve shows WHEN the load occurs; the LDC shows HOW LONG each load lasts.

6. Definitions and Factors

Basic definitions

- ***Connected load:*** sum of the continuous ratings of all the equipment connected to the system (kW).
- ***Maximum demand (MD):*** the greatest demand of load on the station during a given period (usually 15 or 30 min average).

- **Average load (demand):** units generated in a period / hours in the period.
- **Installed capacity:** sum of the ratings of all generators in the station.
- **Plant capacity / Reserve:** Reserve capacity = installed capacity - peak load.

Factor	Definition	Formula
Demand factor	Max demand / connected load (always < 1)	MD / Connected load
Load factor	Average load / maximum demand	Units generated / (MD x hours)
Diversity factor	Sum of individual max demands / simultaneous max demand of the group (always >= 1)	Sum of MD_i / MD_{group}
Plant capacity factor	Actual energy produced / (installed capacity x hours)	Avg load / installed capacity
Plant use factor	Actual energy / (capacity x hours the plant was operating)	Energy / (cap x hrs in use)
Utilisation factor	Max demand on the plant / plant capacity	MD / capacity

- **Load factor** is high for a good, economical plant -> cost per unit falls because fixed charges are spread over more units.
- **Diversity factor** high -> less installed capacity needed for the same consumers -> **lower cost** of supply.

* Always remember: Load factor <= 1, Demand factor <= 1, Diversity factor >= 1, Capacity factor <= Load factor (for a plant with reserve).

Relationship

Plant capacity factor = Load factor x (Max demand / Installed capacity) = Load factor x Utilisation factor

- **Reserve** : hot reserve (running, ready), spinning reserve (synchronised, running), cold reserve (stand-by, can be started on demand).

7. Tariff (Cost of Electrical Energy)

- **Tariff** = the rate at which electrical energy is sold to consumers. A good tariff should **recover** fixed + running + profit, be simple, and be fair to different consumers.
- Total cost is divided into: (a) **fixed cost** (depends on max demand / capacity), (b) **semi-fixed** (depends on connected load / customers), (c) **running cost** (depends on kWh).

Types of tariffs

Type	Expression	Remarks
Flat rate	Rs x per kWh	Same for all consumers; simple, unfair
Block rate	First 100 units at Rs a, next at Rs b	Decreasing price with consumption
Two-part	Rs/year = a x kW + b x kWh	a = fixed (kW), b = running; **Hopkinson** tariff
Maximum demand tariff	Charge on MD + energy used	Needs a MD meter; for industries
Three-part	Rs = a + b x kW + c x kWh	a = fixed per customer (billing, meter reading)
Power factor tariff	Rebate / penalty with p.f.	Encourages high p.f. (>= 0.9)
Time-of-day (ToD)	Higher rate at peak hours	Reduces peak, flattens load curve
Wright / Doherty	Wright: kW x connected-load basis; Doherty: tariff by use	Rarely used now

- Low power factor -> more current for the same power -> bigger cables / transformers, more loss; so utilities charge a **penalty**.

8. Related Exercises (worked)

Example 1: Load factor and capacity factor

- A plant of **installed capacity 100 MW** produces **4.9 x 10^8 kWh** in a year. Maximum demand = **80 MW**.
- Average load = 4.9 x 10^8 / 8760 = **55.9 MW**
- Load factor = 55.9 / 80 = **0.70 (70%)**; Capacity factor = 55.9 / 100 = **0.559 (55.9%)**

Example 2: Diversity factor

- A station supplies loads with maximum demands 8 MW, 10 MW, 12 MW and 15 MW. The simultaneous maximum demand of the station is 30 MW.
- Diversity factor = (8 + 10 + 12 + 15) / 30 = 45 / 30 = **1.5**

Example 3: Daily units and load factor

- A power station has a **daily load**: 0–6 h : 20 MW, 6–12 h : 60 MW, 12–18 h : 40 MW, 18–24 h : 80 MW.
- Energy / day = 6(20 + 60 + 40 + 80) = 6 x 200 = **1200 MWh**; Average load = 1200 / 24 = **50 MW**
- Max demand = 80 MW -> Load factor = 50 / 80 = **0.625 = 62.5%**

Example 4: Two-part tariff

- A consumer has a MD of 100 kW and a load factor of 40%. Tariff: Rs 200 per kW of MD per year + Rs 3 per kWh.
- Energy per year = 100 x 0.4 x 8760 = **3,50,400 kWh**
- Annual bill = 200 x 100 + 3 x 3,50,400 = 20,000 + 10,51,200 = **Rs 10,71,200**; Average rate = Rs 3.06 per kWh

Example 5: Depreciation (straight line)

- A boiler costs Rs 50 lakh, salvage value Rs 5 lakh, life 15 years. Annual depreciation = (50 − 5)/15 = **Rs 3 lakh per year**.

Base-load and peak-load stations

- **Base-load plants** run continuously at high load factor: nuclear, large coal / lignite, run-of-river hydro. Low running cost, slow to start / change load.
- **Peak-load plants** run only a few hours a day: diesel, gas turbine, pumped storage, storage hydro. Quick start (minutes), high running cost.

- **Intermediate / cycling plants:** medium coal and combined-cycle units following the daily variation.
- **Interconnected (grid) operation** gives: better load factor and diversity, less reserve capacity, higher reliability and lower cost per unit; the most efficient plants are loaded first (**merit order dispatch**).

Example 6: Choosing plant capacity

- Annual peak demand of a city is 120 MW, load factor 55%, the reserve required is 15% of peak. Find units supplied per year and installed capacity.
- Energy = 120 x 0.55 x 8760 = **5,78,160 MWh (5.78×10^8 kWh)**
- Installed capacity = 120 x 1.15 = **138 MW**; Capacity factor = 66 / 138 = **47.8%**

Example 7: Annual cost per kWh

- Plant 50 MW, capital cost Rs 1000 crore @ interest + depreciation 12% p.a.; load factor 60%, running cost Rs 2.5 per kWh.
- Energy = 50 x 1000 x 0.6 x 8760 = **26.28×10^7 kWh**; Fixed charges = 0.12 x 1000 crore = Rs 120 crore
- Fixed cost per kWh = $120 \times 10^7 / 26.28 \times 10^7$ = **Rs 4.57**; Total = 4.57 + 2.5 = **Rs 7.07 / kWh**
- If load factor rises to 80%, fixed cost falls to Rs 3.42 -- shows **why a high load factor reduces cost**.

9. Effluents from Power Plants and Impact on Environment

- Every power plant gives out **effluents** (solid, liquid, gaseous, thermal, radioactive) -- need control to protect health and nature.

Plant	Main effluents	Impact
Thermal (coal)	Fly ash, bottom ash, SO_2, NO_x, CO_2, CO, particulates (SPM), hot water	Acid rain, smog, global warming, respiratory disease, thermal pollution of rivers
Diesel / gas	NO_x, CO, unburnt HC, soot, noise	Smog, ozone formation, hearing loss

Plant	Main effluents	Impact
Hydro	No emissions; reservoir submergence	Loss of land and forest, displacement, silting, flora / fauna change, methane from rotting plants
Nuclear	Radioactive waste, hot water	Radiation hazards (cancer, mutation); long-lived waste disposal

Environmental effects in short

- **Greenhouse effect / global warming:** CO_2, CH_4, N_2O trap heat -> sea level rise, climate change.
- **Acid rain:** SO_2 + NO_x + water -> H_2SO_4 and HNO_3; damages crops, buildings, lakes.
- **Particulate matter:** PM_{10} and $PM_{2.5}$ penetrate lungs; fly ash dumping occupies land.
- **Thermal pollution:** warm condenser water discharged raises river temperature -> dissolved oxygen drops -> aquatic life dies.
- **Ozone layer depletion** and **photochemical smog** due to NO_x and HC.

10. Pollutants and Pollution Standards

Types of pollutants

- **Primary pollutants:** emitted directly -- CO, SO_2, NO, particulates, hydrocarbons.
- **Secondary pollutants:** formed in the atmosphere -- ozone, PAN, acid mist, sulphuric acid.

- In India, standards are set by the **CPCB** (Central Pollution Control Board) / State PCBs under the Air (Prevention and Control of Pollution) Act 1981, Water Act 1974 and Environment (Protection) Act 1986.

Parameter	Typical standard / limit
Particulate matter (new coal plants)	30 mg / Nm^3 (revised) ; older units 50–150 mg/Nm^3

Parameter	Typical standard / limit
SO_2	200 mg/Nm^3 (units up to FGD norms) ; stack height 220 m for 500 MW and above
NO_x	300 mg/Nm^3
Mercury	0.03 mg/Nm^3
Condenser cooling water discharge	Temperature rise not more than 5 deg C above intake
Ambient air (NAAQS) SO_2	80 micro-g/m^3 (24 hr) ; PM_{10} : 100 ; $PM_{2.5}$: 60

* Limits are revised from time to time (2015, 2018 notifications) -- remember the idea: ***tall stacks + particulate removal + FGD + low-NO_x burners***.

II. Methods of Pollution Control

Control of particulates

- ***Mechanical collectors / cyclone separators*** -- for coarse dust.
- ***Electrostatic precipitators (ESP)*** -- charge dust particles, collect on plates; efficiency ***99%+***.
- ***Bag (fabric) filters*** and ***wet scrubbers*** -- fine particles.

Control of gaseous emissions

- ***SO_2***: low-sulphur coal, coal washing, ***flue gas desulphurisation (FGD)*** with limestone slurry ($CaCO_3$ + SO_2 -> $CaSO_3$ -> gypsum), fluidised bed combustion with limestone.
- ***NO_x***: low-NO_x burners, staged / lower-temperature combustion, flue-gas recirculation, ***selective catalytic reduction (SCR)*** with ammonia.
- ***CO_2***: higher plant efficiency (supercritical, IGCC), carbon capture, afforestation, renewables.
- ***Tall stacks*** disperse gases over a wide area (dilution, not removal).

Control of water and thermal pollution

- ***Cooling towers*** / cooling ponds instead of once-through cooling; treat effluent and ash-pond overflow (neutralise, settle).

- **Ash:** use fly ash in cement, bricks, road embankments -- also dry ash handling.

Radioactive waste

- Store the spent fuel in water pools -> reprocess -> vitrify and bury in deep geological repositories. Shielding and monitoring with dosimeters.

Noise

- Silencers, acoustic enclosures for diesel sets, ear-muffs for workers.

12. Inspection and Safety Regulations

- Governed by the **Indian Electricity Act 2003**, **Indian Boiler Act 1923** / Indian Boiler Regulations (IBR), the **Factories Act 1948**, and CEA safety regulations.
- **Inspection:** boilers, pressure vessels and pipes are inspected by the **Chief Inspector of Boilers** -- hydraulic test at 1.5 x working pressure, periodic (yearly) inspection and certificate before use.
- Safety valves, water-level gauges, pressure gauges, fusible plugs, feed check valves and blow-down valves must be fitted and tested.

Safety rules in a power plant

- Earthing of all equipment; **lock-out / tag-out** (permit to work) before maintenance.
- Fire protection: fire hydrants, CO_2 / foam extinguishers, **fire walls** and deluge systems for transformers, oil storage and coal bunkers (spontaneous combustion).
- Personal protective equipment: helmets, gloves, goggles, ear plugs, safety shoes.
- Proper ventilation, safe handling of coal dust (explosion risk) and hazardous chemicals (hydrogen, chlorine).
- Emergency shut-down procedures, trained staff, first aid and regular **mock drills**; log books and records kept up to date.
- Nuclear plants: **AERB** (Atomic Energy Regulatory Board) licensing, radiation zones, dose limits and interlocks.

Quick revision

- Load factor = Avg load / Max demand. Demand factor = MD / Connected load. Diversity factor = Sum of MD / Group MD.
- Fixed charges: interest, depreciation, taxes. Running charges: fuel, labour, maintenance.
- Two-part tariff = Rs/kW + Rs/kWh. ESP removes dust; FGD removes SO_2; SCR removes NO_x.